Important Constants

$\mu_\oplus = 398600.5 \frac{km^3}{s^2}$ $\quad R_\oplus = 6378.137\ km$

$G = 6.67259\ x10^{-11}\ \frac{N\,m^2}{kg^2}$ $\quad 1\ N = 1 \frac{kg\,m}{s^2}$

$1W = 1\frac{J}{s}$ $\quad 1J = 1\frac{kg\,m^2}{s^2} = 1Nm$

Restricted 2-Body EOM

$F_g = \frac{Gm_1m_2}{R^2}$ $\quad \ddot{\vec{R}} + \frac{\mu}{R^3}\vec{R} = \ddot{\vec{R}} + \frac{\mu}{R^2}\hat{R} = \vec{0}$

$R = h + R_\oplus$ $\quad h = altitude$

$R = \frac{a(1-e^2)}{1+e\ \cos\ \nu}$ $\quad a = \frac{R_a + R_p}{2}$

$R_a = a(1+e)$ $\quad R_p = a(1-e)$

$\varepsilon = KE + PE = \frac{V^2}{2} - \frac{\mu}{R} = -\frac{\mu}{2a}$

$Period = \mathbb{P} = 2\pi\sqrt{\frac{a^3}{\mu}}$

$V = \sqrt{2\left(\frac{\mu}{R} + \varepsilon\right)} = \sqrt{2\left(\frac{\mu}{R} - \frac{\mu}{2a}\right)}$ $\quad V_{circular} = \sqrt{\frac{\mu}{R}}$

Classical Orbital Elements (COEs)

$\vec{h} = \vec{R} \times \vec{V}$ $\quad \vec{n} = \hat{k} \times \vec{h}$

$\vec{e} = \frac{1}{\mu}\left[\left(V^2 - \frac{\mu}{R}\right)\vec{R} - (\vec{R}\cdot\vec{V})\vec{V}\right]$ $\quad e = |\vec{e}|$

$\cos i = \frac{\hat{k}\cdot\vec{h}}{k\,h}$

$\cos\Omega = \frac{\hat{\imath}\cdot\vec{n}}{i\,n}$ $\quad (if\ n_j < 0, then\ 180° < \ \Omega < 360°)$

$\cos\omega = \frac{\vec{n}\cdot\vec{e}}{n\,e}$ $\quad (if\ e_k < 0, then\ 180° < \ \omega < 360°)$

$\cos\nu = \frac{\vec{e}\cdot\vec{R}}{e\,R}$ $\quad (if\ \vec{R}\cdot\vec{V} < 0, then\ 180° < \ \nu < 360°)$

Alternate COEs

Circular Orbits

$\cos u = \frac{\vec{n}\cdot\vec{R}}{n\,R}$ $\quad (if\ R_k < 0, then\ 180° < \ u < 360°)$

Equatorial Orbits

$\cos\Pi = \frac{\hat{\imath}\cdot\vec{e}}{i\,e}$ $\quad (if\ e_j < 0, then\ 180° < \ \Pi < 360°)$

Circular Equatorial Orbits

$\cos\ell = \frac{\hat{\imath}\cdot\vec{R}}{i\,R}$ $\quad (if\ R_j < 0, then\ 180° < \ \ell < 360°)$

Ground Tracks

For direct orbits, $\mathbb{P} = \frac{\Delta N}{15°/hr}$

$$a = \sqrt[3]{\mu\left(\frac{\mathbb{P}(in\ seconds)}{2\pi}\right)^2}$$

Orbit Prediction $(Keep\ in\ radians)$

$\cos E = \frac{e+\cos\nu}{1+e\cos\nu}$ $\quad \cos\nu = \frac{\cos E - e}{1 - e\cos E}$

$n = \sqrt{\frac{\mu}{a^3}} = \frac{2\pi}{\mathbb{P}}$

$M = E - e\ \sin E$ $\quad 0 \le M < 2\pi$

$M_f - M_i = n\ (TOF) - 2k\pi$

$k \equiv number\ of\ times\ s/c\ passed\ perigee$

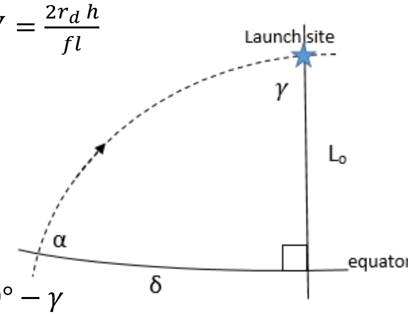
Remote Sensing

$c = 3.0\ x\ 10^8\ m/s$

$\lambda = \frac{c}{f}$ $\quad \lambda_{max} = \frac{2898}{T\ (in\ K)}\ (in\ \mu m)$

$Resolution = \frac{2.44\,\lambda\,h}{D}$ $\quad SW = \frac{2r_d\,h}{fl}$

$FOV = 2\tan^{-1}\frac{r_d}{fl}$



Launch Windows

For direct orbits, $\alpha = i$

$\sin\gamma = \frac{\cos\alpha}{\cos L_o}$

$\beta_{AN} = \gamma$ $\quad \beta_{DN} = 180° - \gamma$

$\cos\delta = \frac{\cos\gamma}{\sin\alpha}$

$LWST_{AN} = \Omega + \delta$ $\quad LWST_{DN} = \Omega + 180° - \delta$

$Wait\ Time = LWST - LST$

Launch Velocities

$$\begin{bmatrix}\Delta V_{needed_S}\\ \Delta V_{needed_E}\\ \Delta V_{needed_Z}\end{bmatrix} = \begin{bmatrix}-V_{bo}\cos\varphi\cos\beta\\ V_{bo}\cos\varphi\sin\beta\\ V_{bo}\sin\varphi\end{bmatrix} - \begin{bmatrix}0\\ V_{launch\ site}\\ 0\end{bmatrix} + \begin{bmatrix}0\\ 0\\ V_{PE\ Gain}\end{bmatrix}$$

$V_{launch\ site} = 0.4651\cos L_o$

$V_{PE\ Gain} = \sqrt{\frac{2\mu\,(R_{bo} - R_{launch\ site})}{R_{bo}\,R_{launch\ site}}}$

$\Delta V_{design} = \left|\Delta\vec{V}_{needed}\right| + V_{losses\ other\ than\ gravity}$

Propulsion

$g_o = 9.81\ m/s^2$

$F_{thrust} = \dot{m}V_{exit} + A_{exit}(P_{exit} - P_{atm})$

$\Delta V = I_{sp}g_o\ln\frac{m_{initial}}{m_{final}}$ $\quad I_{sp} = \frac{F}{\dot{m}g_o}$

Hohmann Transfers

$a_t = \frac{R_1 + R_2}{2}$ $\quad TOF_{Hohmann} = \pi\sqrt{\frac{a_t^3}{\mu}}$

$\Delta V = |V_{orbit} - V_{t\ @\ orbit}|$ $\quad \Delta V_{total} = \Delta V_1 + \Delta V_2$

Plane Changes

$\Delta V_s = 2V_{orbit}\sin\frac{\theta}{2}$ $\quad \theta = change\ in\ \measuredangle$

$\Delta V_c = \sqrt{V_{before}^2 + V_{after}^2 - 2V_{before}V_{after}\cos\theta}$

Rendezvous

$$\omega = \sqrt{\frac{\mu}{R^3_{circular\ orbit}}}$$

$\varphi_{initial} = \measuredangle\ from\ interceptor\ to\ target\ in\ direction\ of\ motion$

Coplanar Rendezvous

$\alpha_{lead} = \omega_{tgt}\ x\ TOF_{Hohmann} \qquad 0 \le \alpha_{lead} < 2\pi$

$\varphi_{final} = \pi - \alpha_{lead}$

$W.T. = \frac{\varphi_{final} - \varphi_{initial}(\pm 2\pi)}{\omega_{tgt} - \omega_{interceptor}} \qquad only\ use\ (\pm 2\pi)\ if\ WT\ is\ negative$

Co-orbital Rendezvous

$If\ target\ ahead, \varphi_{travel} = 2\pi - \varphi_{initial}$

$If\ target\ behind, \varphi_{travel} = 4\pi - \varphi_{initial}$

$$a_{phasing} = \sqrt[3]{\mu \left(\frac{\varphi_{travel}}{2\pi\ \omega_{tgt}}\right)^2}$$

Attitude Control

$D = h\,\psi \qquad \vec{H} = I\vec{\Omega} \qquad \vec{T} = \vec{R} \times \vec{F}$

Electrical Power

$Solar\ Input = 1358 \frac{W}{m^2}$

$\rho = \sin^{-1} \frac{R_\oplus}{R_\oplus + h}$

$T_{eclipse} = \frac{2\rho}{360°}\ x\ \mathbb{P} \qquad T_{sun} = \mathbb{P} - T_{eclipse}$

$E_{eclipse} = (P_{payload} + P_{bus})T_{eclipse}$

$E_{sun} = (P_{payload} + P_{bus})T_{sun}$

$P_{solar\ req'd} = \frac{E_{eclipse} + E_{sun}}{T_{sun}}$

$P_{BOL} = (Solar\ Input)\ \eta\ A_{effective}$

$P_{EOL} = P_{BOL}(1 - DG_{array})^{msn\ life}$

$Cap_{Batt\ Req'd} = \frac{E_{eclipse}}{DoD}$

Thermal Control

$\tau + \alpha + \rho = 1$

$Solar\ Input = 1358 \frac{W}{m^2}$

$Albedo = 407 \frac{W}{m^2} \qquad Earthshine = 237 \frac{W}{m^2}$

$q_{in} = (Heat\ Input) A\ \alpha$

$q_{int} = P_{payload} + P_{bus} \qquad q_{out} = q_{in} + q_{int}$

$q_{out} = \sigma \varepsilon_{total} A\, T^4 \qquad \sigma = 5.67\ x\ 10^{-8} \frac{W}{m^2 K^4}$

$\varepsilon_{total} = \frac{q_{out}}{\sigma A T^4} \qquad T = \sqrt[4]{\frac{q_{out}}{\sigma\ A\ \varepsilon_{total}}}$

Structures

$f_{fund} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad \sigma = \frac{F}{A} \qquad \varepsilon = \frac{\Delta l}{l}$

Unit Conversions

Temperature: $K = 273.15 + °C \qquad °C = \frac{5}{9}(°F - 32)$

Angular distance: π radians = 180°

Communications: 1 Byte = 8 bits

Binary prefixes: $ki = 2^{10} \quad Mi = 2^{20} \quad Gi = 2^{30}$

SI prefixes: $k = 10^3 \quad M = 10^6 \quad G = 10^9 \quad \mu = 10^{-6} \quad n = 10^{-9}$

Communications

$\lambda = \frac{c}{f} \qquad c = 3.0\ x\ 10^8\ m/s$

$$\frac{S}{N} = \left(\frac{P_{xmtr} G_{xmtr}}{k\ B}\right)\left(\frac{\lambda}{4\pi R}\right)^2\left(\frac{G_{rcvr}}{T_{rcvr}}\right) = \frac{P_{xmtr} G_{xmtr} \left(\frac{\lambda}{4\pi R}\right)^2 G_{rcvr}}{k\ T_{rcvr}\ B}$$

$R_{max} = \sqrt{(R_\oplus + h)^2 - {R_\oplus}^2} \qquad k = 1.381\ x 10^{-23} \frac{J}{K}$

Re-entry

$\gamma \equiv reentry\ angle$

$\beta = 0.000139\ m^{-1} \qquad \rho_o = 1.225 \frac{kg}{m^3}$

$BC = \frac{m}{C_D A} \qquad a_{drag} = \frac{\rho V^2}{2\ BC}$

$a_{drag_{max}} = \frac{V^2_{reentry}\,\beta \sin\gamma}{2e} \qquad \dot{q} = 1.83 x 10^{-4} V^3 \sqrt{\frac{\rho}{r_{nose}}}$

$h_{a_{drag max}} = \frac{1}{\beta} \ln \frac{\rho_o}{BC\beta \sin\gamma} \qquad h_{\dot{q}_{max}} = \frac{1}{\beta} \ln \frac{\rho_o}{3BC\beta \sin\gamma}$

$V_{\dot{q}_{max}} = 0.846\ V_{reentry}$

Ballistic Missiles

$\Lambda \equiv range\ angle\ (in\ deg) \qquad \beta \equiv azimuth\ (in\ deg)$

$\cos\Lambda = \sin L_o \sin L_t + \cos L_o \cos L_t \cos\Delta\ell$

$\Delta\ell = \ell_t - \ell_o \qquad \cos\beta = \frac{\sin L_t - \sin L_o \cos\Lambda}{\cos L_o \sin\Lambda}$

$Q_{bo} = \frac{V^2_{bo} R_{bo}}{\mu} \qquad Q_{bo_{min}} = \frac{2 \sin\frac{\Lambda}{2}}{1 + \sin\frac{\Lambda}{2}}$

$$\varphi_{bo_{low}} = \frac{1}{2}\left\{\sin^{-1}\left[\left(\frac{2 - Q_{bo}}{Q_{bo}}\right)\sin\frac{\Lambda}{2}\right] - \frac{\Lambda}{2}\right\}$$

$$\varphi_{bo_{high}} = \frac{1}{2}\left\{180° - \sin^{-1}\left[\left(\frac{2 - Q_{bo}}{Q_{bo}}\right)\sin\frac{\Lambda}{2}\right] - \frac{\Lambda}{2}\right\}$$

$\Lambda_{max} = 2\sin^{-1}\frac{Q_{bo}}{2 - Q_{bo}} \qquad \varphi_{bo_{\Lambda\,max}} = 45° - \frac{\Lambda_{max}}{4}$

Rotating Earth Correction

$\cos\Lambda = \sin L_o \sin L_t + \cos L_o \cos L_t \cos(\Delta\ell + \omega_{Earth} TOF)$

Interplanetary Travel

$\mu_{sun} = 1.327 x 10^{11} \frac{km^3}{s^2}$

$V_\infty = |V_{planet} - V_{transfer@planet}| \qquad \varepsilon_\infty = \frac{V^2_\infty}{2}$

$V_{hyperbolic} = \sqrt{2\left(\frac{\mu_{planet}}{R} + \varepsilon_\infty\right)}$

$\Delta V = |V_{park} - V_{hyperbolic}|$